

Imran Jahid Khan

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Research Interests

Cyber-Physical Systems • Reinforcement Learning • AI Security & Robustness • Cybersecurity • Cloud-Based Control Systems • Resilient Autonomous Systems

Education

Tuskegee University, Alabama, USA

Aug 2025 – May 2027 (expected)

M.S. in Information Systems and Computer Security

Bangladesh University of Engineering & Technology (BUET), Dhaka

2011 – 2016

B.Sc. in Mechanical Engineering

Publications

PEER-REVIEWED

1. M. A. Rahman, I. J. Khan, and F. Wu, "Detection of Integrity Attacks in Cloud-Based Cyber-Physical Systems (CBCPS) Using Long Short-Term Memory (LSTM) Models," in *Proc. 2026 9th International Conference on Information and Computer Technologies (ICICT)*, pp. 533–538, 2026.

ACCEPTED & FORTHCOMING

2. I. J. Khan, M. Hossain, M. A. Rahman, and F. Wu, "Resilience Analysis of Cloud-Controlled Cyber-Physical Systems Using a Reinforcement Learning CartPole Testbed," *AllIoT Conference*, 2026. (Accepted; presented at the conference in Seattle, WA.)

UNDER REVIEW

3. I. J. Khan, M. Hossain, M. A. Rahman, and F. Wu, "Blockchain-Anchored Detection and Recovery for Resilient Reinforcement Learning-Based Cyber-Physical Systems," under review, *7th Annual National HBCU Blockchain, FinTech, and AI Conference*, Morgan State University, 2026.
4. I. J. Khan, M. Hossain, M. A. Rahman, and F. Wu, "Reconstruction-Assisted DQN for Resilient Cloud-Controlled Cyber-Physical Systems against Data Integrity Attacks," under review, *ICMLA*, 2026.
5. M. Hossain, I. J. Khan, M. A. Rahman, and F. Wu, "Evaluating Learning Efficiency and Computational Trade-Offs in Classical and Quantum Reinforcement Learning Using a CartPole Testbed," under review, *ICMLA*, 2026.

Honors & Scholarships

- **Full Scholarship (Graduate Assistantship)**, M.S. in Information Systems and Computer Security, Tuskegee University – 2025 (accepted)
- **Japan Human Resources Development Scholarship (JDS)**, Government of Japan, for graduate study at Hiroshima University – 2025 (declined)
- **Full Scholarship**, M.A. in Policy Economics, Williams College – 2025 (declined)
- **Prime Minister's Fellowship**, Government of Bangladesh – 2024
- **Foreign Master's Scholarship**, "Capacity Building of BCS Administration Academy (2nd Revised)" Project, Government of Bangladesh – 2024

Research Experience

Graduate Research Assistant

Aug 2025 – present

Dept. of Computer Science, Tuskegee University • Advisor: Dr. Mohammad Arafatur Rahman

- Research the security and resilience of cloud-based cyber-physical systems, focusing on learning-based control under adversarial and noisy conditions.
- Developed LSTM-based detection of controller integrity and replay attacks in cloud-based CPS.
- Built a reinforcement-learning CartPole testbed (DQN) for resilience analysis and reconstruction-assisted recovery against data-integrity attacks; studied classical vs. quantum RL efficiency trade-offs.

- Impact of a Housing Project on Children's Lifestyle (2023).
- Wind Turbine Blade Design for Optimum Power Output, B.Sc. thesis, BUET (2016).

Teaching Experience

Graduate Teaching Assistant

Aug 2025 – present

Introduction to Computer Science, Dept. of Computer Science, Tuskegee University

Presentations

- Poster, AWS-MLU AI/ML Fall Teaching and Research Symposium & Student Poster Session, 2025.

Professional Experience (Bangladesh Civil Service)

Deputy Director (Senior Assistant Secretary)

Sep 2024 – Jul 2025

Export Promotion Bureau, Ministry of Commerce

- Conducted export-policy analysis with data-driven recommendations and digitized trade-facilitation processes with stakeholders.

Assistant Commissioner (Land) & Executive Magistrate

Nov 2022 – Aug 2024

Upazila Land Office, Patuakhali & Bhola, Ministry of Land

- Managed digital public-land records and data-driven monitoring; adjudicated disputes and coordinated field administration and digital reporting.

Assistant Commissioner & Executive Magistrate

Apr 2019 – Nov 2022

Office of the Deputy Commissioner, Cox's Bazar, Ministry of Public Administration

- Oversaw tourism facilitation, budget/HR management, and e-procurement; led disaster-relief and COVID-19 response using ICT for coordination.

Technical Skills

Programming: Python, C, C++

ML / Methods: Reinforcement Learning (DQN), Deep Learning (LSTM), data-driven anomaly detection

Tools: SolidWorks, SPSS, MS Office, Google Workspace, $\text{\LaTeX}$

Standardized Tests

GRE: 322 (V 155, Q 167, AW 3.5) – Dec 2024

IELTS Academic: 7.0 (L 8.0, R 7.5, W 6.5, S 6.5) – Jan 2024

Professional Training

Law & Administration, Public Procurement, Land Management, and Government Research – BPATC, BCSAA, LATC, and BIGM (2019–2025).